

PAPER_963: Buoyancy Gravity Triadic Mode

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 215 **Source:** triadic_solutions_next.py (BuoyancyGravityTriadic) **Calculator:** BuoyancyGravityTriadicCalc (CP4 #547) **CVW:** v2.0.0 compliant

Abstract

The Buoyancy Gravity Triadic mode evaluates $E_{\text{net}}(t, \Gamma) = S_{26} \cdot \cos(\omega_{\text{SCm}} t) \cdot \exp(-\Gamma t) - \text{threshold}$. Positive E_{net} drives expansion (nebulae, HII regions); negative E_{net} drives erosion (filaments, cometary knots).

1. Net Energy

$$E_{\text{net}}(t, \Gamma) = S_{26} \cdot \cos(\omega_{\text{SCm}} \cdot t) \cdot \exp(-\Gamma \cdot t) - \text{threshold}$$

2. Regime Classification

E_{net}	Regime	Astrophysical Example
> 0.1	Expansion	Nebulae, HII regions
< -0.1	Erosion	Filaments, pillars
$[-0.1, 0.1]$	Neutral	Transition zones

References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
 2. PAPER_961 — Compressed Gravity Triadic
 3. PAPER_962 — Resonant Gravity Triadic
 4. PAPER_966 — Unified Triadic Solver
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Cross-Links

Related Paper	Relationship
PAPER_961	Compressed branch of same triadic
PAPER_962	Resonant branch
PAPER_966	Unified solver combining all three
PAPER_954	$E(t)$ sign-flip from buoyancy dynamics

Calibration Constants

Constant	Symbol	Value	Validation Domain
κ	—	$5.0 \times 10^{-4} \text{ day}^{-1}$	Damping
$[SSq]$	—	0.57	String coupling
β_i	—	0.603	Buoyancy coupling
E_{net} sign	—	+: expansion, —: erosion	Phase indicator

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
E_{net} sign-flip	Expansion $\leftrightarrow$ erosion phase transition	Derived
F_{UBi} buoyancy force	Acts outward against g_{comp}	Validated

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: Buoyancy Gravity (F_{UBi} Net Energy Balance)

§A.2 Lagrangian Density

$$\mathcal{L}_{\text{buoy}} = F_{UBi}(r) \cdot r - \int_0^r g_{\text{comp}}(r') dr'$$

§A.3 Euler-Lagrange Equation of Motion

$$E_{\text{net}}(r) = F_{UBi}(r) \cdot r - \int_0^r g_{\text{comp}}(r') dr', \quad \text{sign}(E_{\text{net}}) : +\text{expand}, -\text{erode}$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 $\rightarrow$ buoyancy force $F_{UBi} \rightarrow$ net energy balance $\rightarrow$ expansion/erosion phase $\rightarrow$ triadic branch 3/3

§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS

F_{UBi} profile traces VDS outward pressure gradient.

§B.2 DVP

Buoyancy vortex dipole: inward gravity vs. outward F_{UBi} .

§B.3 BSH

E_{net} saturates at stellar surface (BSH envelope crossing).

§B.4 Production-Scale Consistency

Metric	Value	Status
β_i	0.603	Confirmed
$[SSq]$	0.57	Confirmed